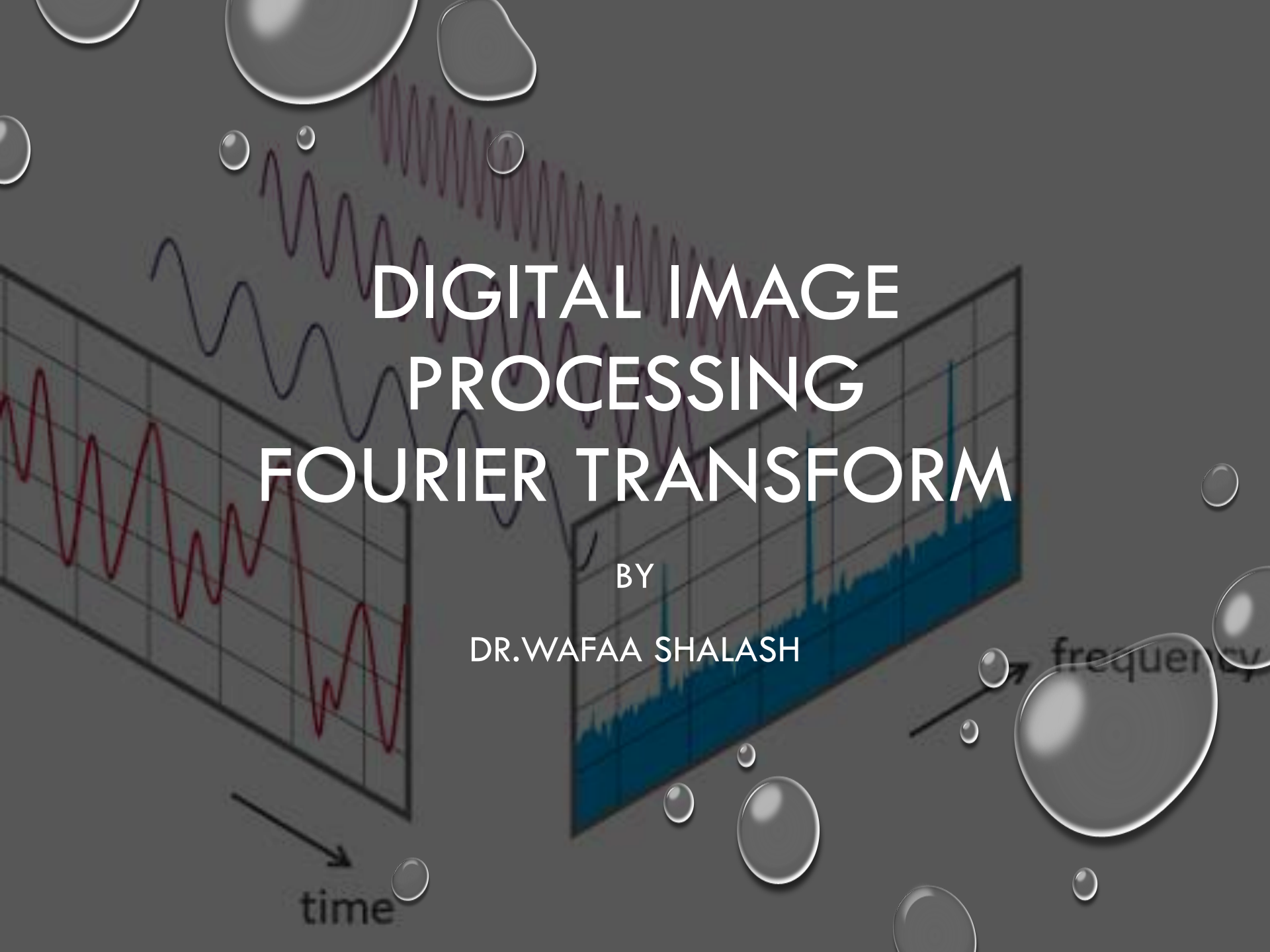
DIGITAL IMAGE PROCESSING FOURIER TRANSFORM

BY

DR.WAFAA SHALASH

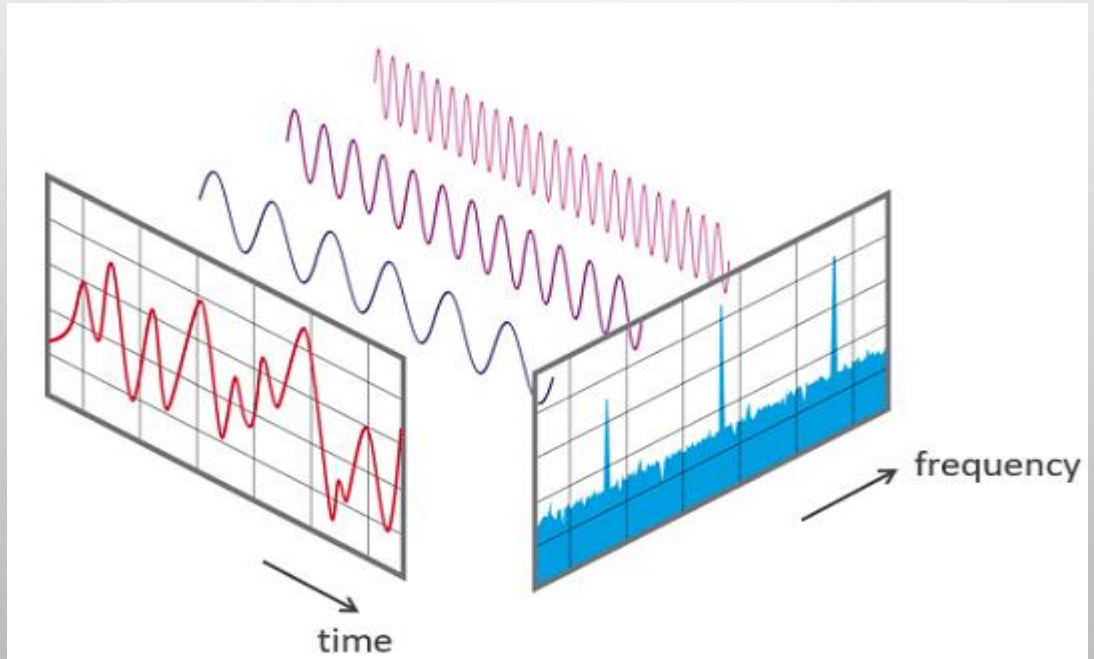


INTRODUCTION

Jean Baptiste Joseph Fourier

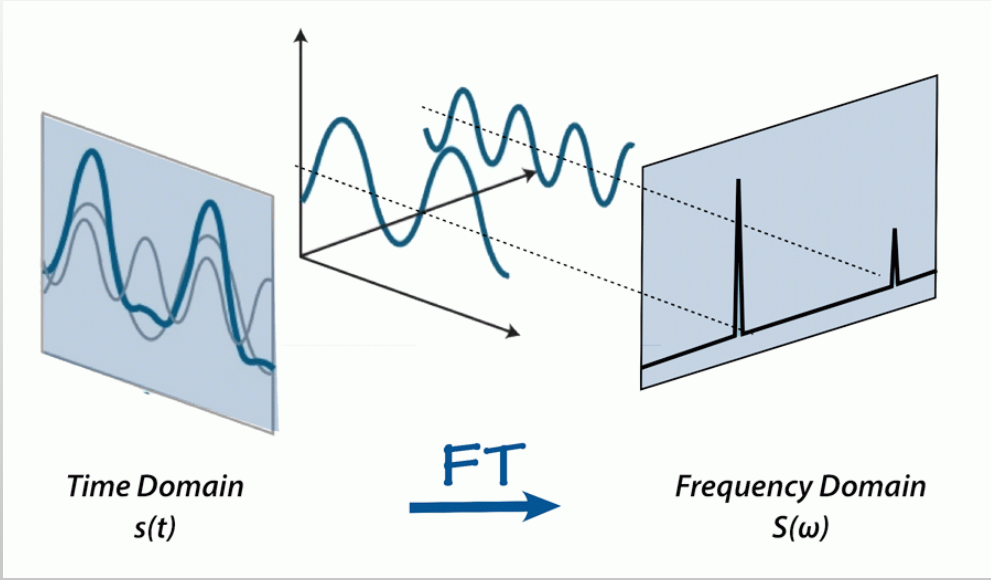


- THE FOURIER TRANSFORM IS GENERALLY USED TO DECOMPOSE A SIGNAL INTO VARIOUS SINUSOIDAL COMPONENTS.
- FOR AN IMAGE, THE OUTPUT OF THE TRANSFORMATION IS THE REPRESENTATION OF THE IMAGE IN FREQUENCY SPACE, WHILE THE INPUT IMAGE IS THE REAL SPACE EQUIVALENT.
- IN THE FOURIER SPACE IMAGE, EACH POINT REPRESENTS A PARTICULAR FREQUENCY CONTAINED IN THE REAL DOMAIN IMAGE.



SIGNIFICANCE

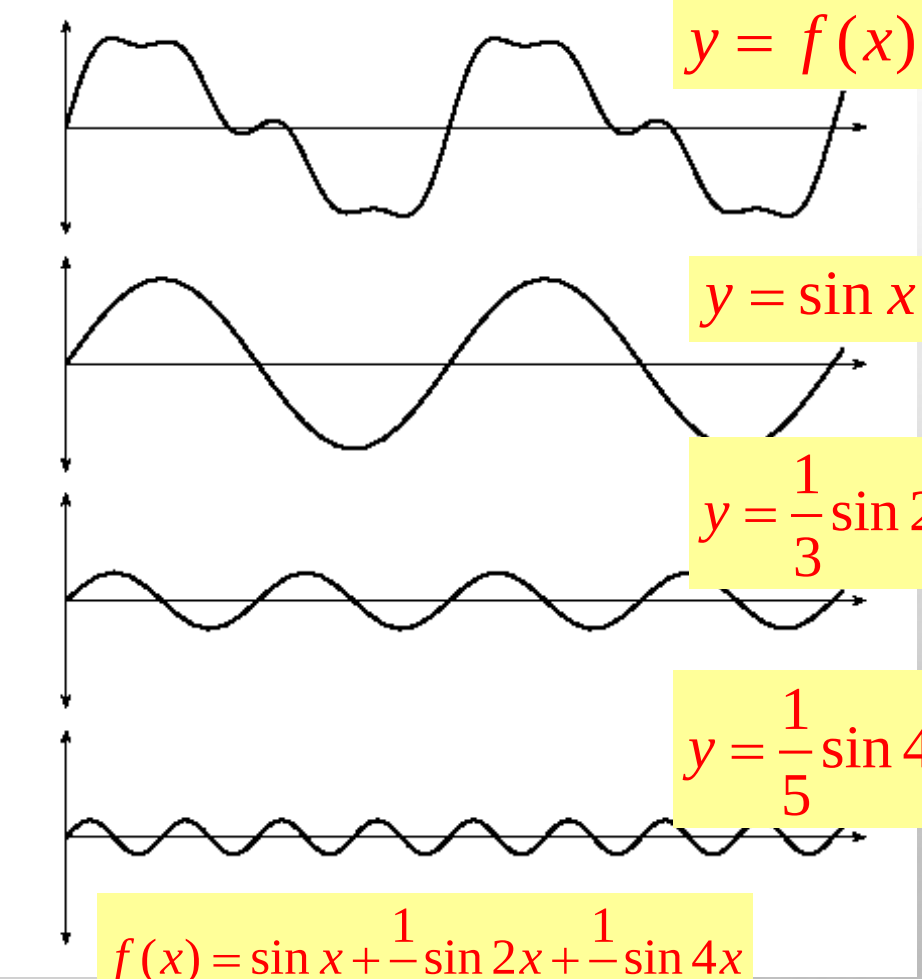
- THE FOURIER TRANSFORM ALLOWS US TO PERFORM TASKS WHICH WOULD BE IMPOSSIBLE TO PERFORM ANY OTHER WAY; ITS EFFICIENCY ALLOWS US TO PERFORM OTHER TASKS MORE QUICKLY.
- THE FOURIER TRANSFORM PROVIDES A POWERFUL ALTERNATIVE TO LINEAR SPATIAL FILTERING; IT IS MORE EFFICIENT TO USE THE FOURIER TRANSFORM THAN A SPATIAL FILTER FOR A LARGE FILTER.
- THE FOURIER TRANSFORM ALSO ALLOWS US TO ISOLATE AND PROCESS PARTICULAR IMAGE FREQUENCIES, AND SO PERFORM LOW-PASS AND HIGH-PASS FILTERING WITH A GREAT DEGREE OF PRECISION.



SOME INTUITION

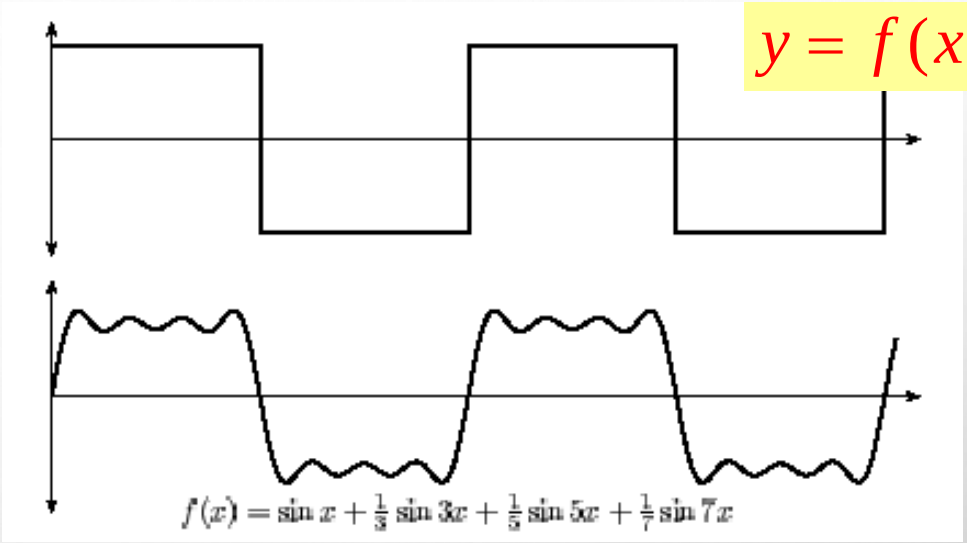
- A PERIODIC FUNCTION MAY BE WRITTEN AS THE SUM OF SINES AND COSINES OF VARYING AMPLITUDES AND FREQUENCIES.

EXAMPLES =>



SOME INTUITION

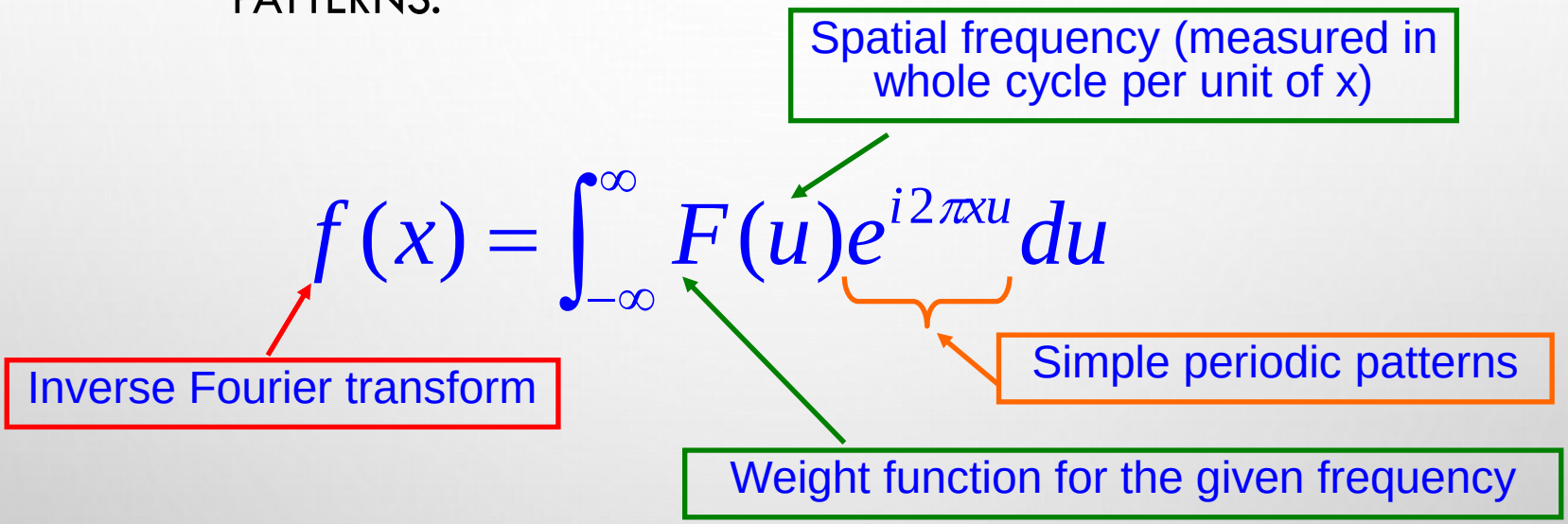
- SOME FUNCTIONS WILL REQUIRE ONLY A FINITE NUMBER OF FUNCTIONS IN THEIR DECOMPOSITION; OTHERS WILL REQUIRE AN INFINITE NUMBER.



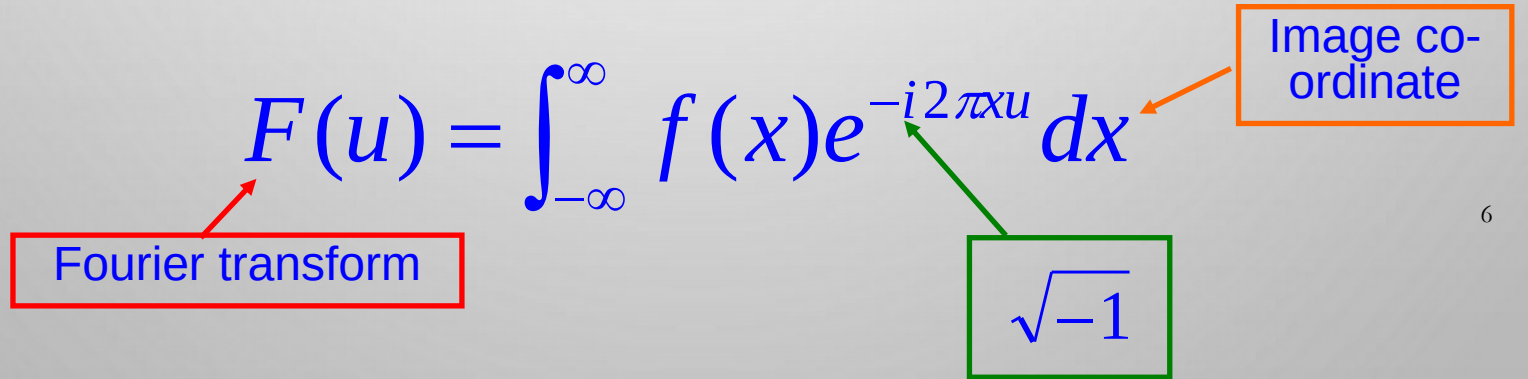
$$f(x) = \sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x + \frac{1}{7}\sin 7x + \frac{1}{9}\sin 9x + \ldots$$

1-D CONTINUOUS

- F(X) IS A LINEAR COMBINATION OF SIMPLE PERIODIC PATTERNS.



- WHERE

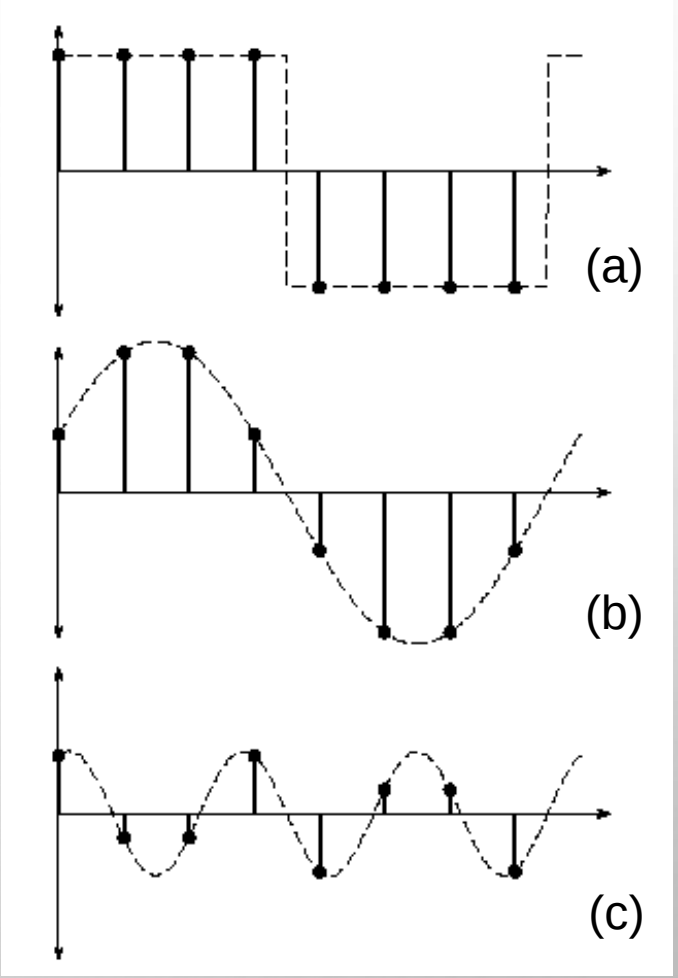


1-D DISCRETE (CON'D)

- IN IMAGE PROCESSING, WE DEAL WITH A DISCRETE FUNCTION.
- SINCE WE ONLY HAVE TO OBTAIN A FINITE NUMBER OF VALUES, WE ONLY NEED A FINITE NUMBER OF FUNCTIONS TO DO IT.
- FOR EXAMPLE: 1 1 1 1 -1 -1 -1 -1,

  WHICH WE MAY TAKE AS A DISCRETE APPROXIMATION TO THE SQUARE WAVE

  OF FIGURE (A). THIS CAN BE EXPRESSED AS THE SUM OF TWO SINE FUNCTIONS, (B) AND (C)



DEFINITION OF 1-D DFT

Suppose $f = [f_0, f_1, f_2, \cdots, f_{N-1}]$

is a sequence of length N. Define its discrete Fourier transform

$$F = [F_0, F_1, F_2, \cdots, F_{N-1}]$$

where $$F_u = \frac{1}{N} \sum_{x=0}^{N-1} \exp[-2\pi i \frac{xu}{N}] \cdot f_x$$

We can express this definition as matrix multiplication

$$F = \mathrm{F} f$$

Where F is an NxN matrix defined by $$\mathrm{F}_{m,n} = \frac{1}{N} \exp[-2\pi i \frac{mn}{N}]$$

DEFINITION OF 1-D DFT

Given N, we shall define $\omega = \exp[\frac{-2\pi i}{N}]$

So that $F_{m,n} = \frac{1}{N}\omega^{mn}$

$$F_{m,n} = \frac{1}{N}\exp[-2\pi i \frac{mn}{N}]$$

Then we can write

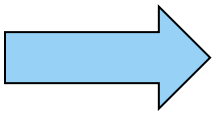
$$F = \frac{1}{N}\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega^{1} & \omega^{2} & \omega^{3} & \omega^{4} & \cdots & \omega^{N-1} \\ 1 & \omega^{2} & \omega^{4} & \omega^{6} & \omega^{8} & \cdots & \omega^{2(N-1)} \\ 1 & \omega^{3} & \omega^{6} & \omega^{9} & \omega^{12} & \cdots & \omega^{3(N-1)} \\ 1 & \omega^{4} & \omega^{8} & \omega^{12} & \omega^{16} & \cdots & \omega^{4(N-1)} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \omega^{2(N-1)} & \omega^{3(N-1)} & \omega^{4(N-1)} & \cdots & \omega^{(N-1)^2} \end{bmatrix}$$

Suppose $f = [1, 2, 3, 4]$ so that N=4. Then

$$\omega = \exp[\frac{-2\pi i}{4}]$$
$$= \exp[\frac{-\pi i}{2}]$$
$$= \cos(-\frac{\pi}{2}) + i\sin(-\frac{\pi}{2})$$
$$= -i$$

Then we have

$$\mathrm{F} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & (-i)^2 & (-i)^3 \\ 1 & (-i)^2 & (-i)^4 & (-i)^6 \\ 1 & (-i)^3 & (-i)^6 & (-i)^9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix}$$



$$\mathrm{F} = \frac{1}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix}\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 10 \\ -2+2i \\ -2 \\ -2-2i \end{bmatrix}$$

$F = \mathrm{F}f$

THE INVERSE DFT

Inverse DFT: $f_x = \sum_{u=0}^{N-1} \exp\left[2\pi i \frac{xu}{N}\right] F_u$

Difference with forward transform:

(1). There is no scaling factor 1/N

(2). The sign inside the exponential function has been changed to positive

Inverse DFT can also be expressed as matrix product

$$f = \mathcal{F}^{-1} F$$

with

$$\mathcal{F}^{-1} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & \cdots & 1 \\ 1 & \overline{w}^1 & \overline{w}^2 & \overline{w}^3 & \overline{w}^4 & \cdots & \overline{w}^{N-1} \\ 1 & \overline{w}^2 & \overline{w}^4 & \overline{w}^6 & \overline{w}^8 & \cdots & \overline{w}^{2(N-1)} \\ 1 & \overline{w}^3 & \overline{w}^6 & \overline{w}^9 & \overline{w}^{12} & \cdots & \overline{w}^{3(N-1)} \\ 1 & \overline{w}^4 & \overline{w}^8 & \overline{w}^{12} & \overline{w}^{16} & \cdots & \overline{w}^{4(N-1)} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \overline{w}^{N-1} & \overline{w}^{2(N-1)} & \overline{w}^{3(N-1)} & \overline{w}^{4(N-1)} & \cdots & \overline{w}^{(N-1)^2} \end{bmatrix}$$

where

$$\overline{w} = \frac{1}{w} = \exp\left[\frac{2\pi i}{N}\right]$$

MATLAB FUNCTIONS: *FFT/IFFT*

- EXAMPLE:

```
» a=[1 2 3 4 5 6 7 8 9]

» b=fft(a)

  45.0          -4.5 +12.3i   -4.5 + 5.3i   -4.5 + 2.5i
 -4.5 + 0.7i   -4.5 - 0.7i   -4.5 - 2.5i   -4.5 - 5.3i
 -4.5 -12.3i

» ifft(b)

  1.0 - 0.0i    2.0 - 0.0i    3.0  - 0.0i    4.0  - 0.0i
  5.0- 0.0i     6.0 - 0.0i    7.0  + 0.0i    8.0  + 0.0i
  9.0 + 0.0i
```

PROPERTIES OF 1-D DFT

- LINEARITY: SUPPOSE **F** AND **G** ARE TWO VECTORS OF SAME LENGTH, AND **P** AND **Q** ARE SCALARS, WITH **H = PF + QG**. IF **F,G** AND **H** ARE THE DFT'S OF **F,G** AND **H**, THEN

$$\mathbf{H = pF + qG}$$

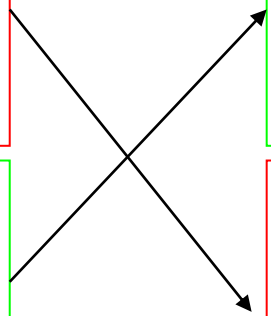
- SHIFTING: SUPPOSE WE MULTIPLY EACH ELEMENT X_N OF A VECTOR X BY $(-1)^N$ I.E., WE CHANGE THE SIGN OF EVERY SECOND ELEMENT. LET THE RESULTING VECTOR BE DENOTED X'. THEN DFT X' OF X' IS EQUAL TO THE DFT X OF X WITH THE SWAPPING OF THE LEFT AND RIGHT HALVES.

Applications of some of the properties are shown in next lecture

```
» x = [ 2   3   4   5   6   7   8   1]
» x1 = (-1).^[0:7].*x
x1 = 2   -3    4   -5    6   -7    8   -1

» X=fft(x')                          » X1=fft(x1')

X =                                  X1 =
  36.0000                              4.0000
  -9.6569 + 4.0000i                    1.6569 + 4.0000i
  -4.0000 - 4.0000i                   -4.0000 + 4.0000i
   1.6569 - 4.0000i                   -9.6569 - 4.0000i
   4.0000                             36.0000
   1.6569 + 4.0000i                   -9.6569 + 4.0000i
  -4.0000 + 4.0000i                   -4.0000 - 4.0000i
  -9.6569 - 4.0000i                    1.6569 - 4.0000i
```



Then the DFT X1 of x1 is equal to the DFT X of x with the swapping of the left and right halves.

PROPERTIES OF 1-D DFT (CON'D)

- CONJUGATE SYMMETRY: IF X IS REAL, AND OF LENGTH N, THEN ITS DFT **X** SATISFIES THE CONDITION $X_k = \overline{X_{N-k}}$, WHERE $\overline{X_{N-k}}$ IS THE COMPLEX CONJUGATE OF X_{N-k} FOR ALL K=1,2,3,…,N-1. (CHECK OUT PREVIOUS SLIDE)
- CIRCULAR CONVOLUTION: SUPPOSE X AND Y ARE TWO VECTORS OF THE SAME LENGTH N. THEN WE DEFINE THEIR CONVOLUTION TO BE THE VECTOR, $z = x * y$, WHERE $z_k = \frac{1}{n}\sum_{n=0}^{N-1} x_n y_{k-n}$

EXAMPLE

$$z_0 = \frac{1}{4}(x_0 y_0 + x_1 y_{-1} + x_2 y_{-2} + x_3 y_{-3})$$

$$z_1 = \frac{1}{4}(x_0 y_1 + x_1 y_0 + x_2 y_{-1} + x_3 y_{-2})$$

$$z_2 = \frac{1}{4}(x_0 y_2 + x_1 y_1 + x_2 y_0 + x_3 y_{-1})$$

$$z_3 = \frac{1}{4}(x_0 y_3 + x_1 y_2 + x_2 y_1 + x_3 y_0)$$

$$\square\!\square\!\square\ y_0\ \ y_1\ \ y_2\ \ y_3\ \ y_0\ \ y_1\ \ y_2\ \ y_3\ \square\!\square\!\square$$

$$=\square\!\square\!\square\ y_0\ \ y_{-3}\ \ y_{-2}\ \ y_{-1}\ \ y_0\ \ y_1\ \ y_2\ \ y_3\ \square\!\square\!\square$$

Thus $y_{-1} = y_3$, $y_{-2} = y_2$, $y_{-3} = y_1$

PROPERTIES OF 1-D DFT (CON'D)

- CIRCULAR CONVOLUTION: CAN BE DEFINED IN TERMS OF POLYNOMIAL PRODUCTS.
  - SUPPOSE **P(U)** THE POLYNOMIAL IN **U** WHOSE COEFFICIENTS ARE ELEMENTS OF **X**. LET **Q(U)** BE THE POLYNOMIAL WHOSE COEFFICIENTS ARE ELEMENTS OF **Y**. FROM THE PRODUCT **P(U)Q(U)(1+U^N)**, AND EXTRACT THE COEFFICIENTS OF **U^N** TO **U^{2N-1}**, THESE WILL BE THE REQUIRED CIRCULAR CONVOLUTION

  - EXAMPLE: $x=[1,2,3,4],\quad y=[5,6,7,8]$

    WE HAVE $p(u)=1+2u+3u^2+4u^3$ AND $q(u)=5+6u+7u^2+8u^3$

    THEN WE EXPAND

$$p(u)q(u)(1+u^4)=5+16u+34u^2+60u^3+66u^4+68u^5+66u^6+60u^7+61u^8+52u^9+32u^{10}$$

EXTRACTING THE COEFFICIENTS OF U4,U5,…., U7 WE OBTAIN

$$x*y=[66,68,66,60]$$

IMPORTANCE OF CONVOLUTION

- SUPPOSE **X** AND **Y** ARE VECTORS OF EQUAL LENGTH. THEN THE DFT OF THEIR *CIRCULAR CONVOLUTION* IS EQUAL TO THE ELEMENT-BY-ELEMENT PRODUCT OF THE DFT'S OF X AND Y.
- IF Z,X,Y ARE THE DFT'S OF Z=X*Y, X AND Y RESPECTIVELY, THEN Z=X.Y

Example:

```
» fft(cconv(a,b)')
ans =

  1.0e+002 *

   2.6000
  -0.0000 - 0.0800i
   0.0400
  -0.0000 + 0.0800i
```

```
» fft(a').*fft(b')
ans =
  1.0e+002 *

   2.6000
  -0.0000 - 0.0800i
   0.0400
  -0.0000 + 0.0800i
```

MORE ON DFT

- IN GENERAL, THE TRANSFORM INTO THE FREQUENCY DOMAIN WILL BE A COMPLEX VALUED FUNCTION, THAT IS, WITH MAGNITUDE AND PHASE.

$$\text{magnitude} = \|F_u\| = \sqrt{F_{real} * F_{real} + F_{imag} * F_{imag}}$$

$$\text{phase} = \tan^{-1}\left(\frac{F_{imag}}{F_{real}}\right)$$

$$F_u = \|F_u\| \exp(i\theta)$$

- THE DC COEFFICIENT: THE VALUE F(0)

$$F(0) = \frac{1}{N}\sum_{x=0}^{N-1} f_x \exp(0) = \frac{1}{N}\sum_{x=0}^{N-1} f_x$$

AVERAGE OF THE INPUT SERIES.

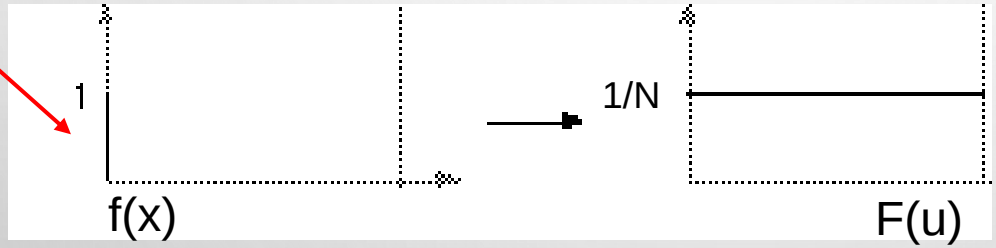
SOME PROPERTIES OF TRANSFORM PAIR

- SCALING RELATIONSHIP: $f\left(\frac{x}{a}\right) \rightarrow aF(au); \quad f(ax) \rightarrow \frac{1}{a}F\left(\frac{u}{a}\right)$
- TIME SHIFT / FREQUENCY MODULATION:

$$f(x+a) \rightarrow F(u)e^{i2\pi au/M}; \quad f(x)e^{i2\pi ax/M} \rightarrow F(u-a)$$

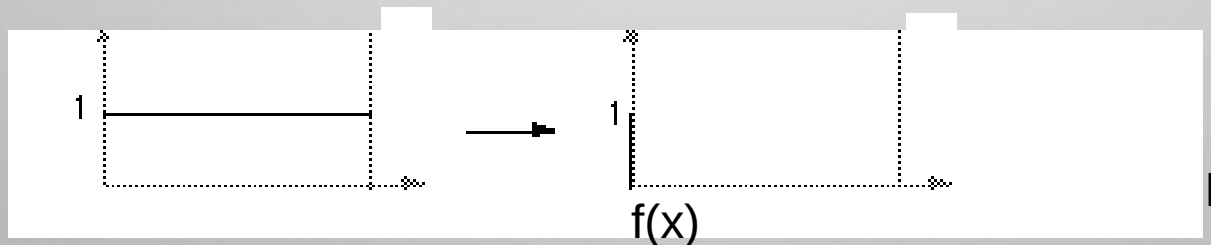
- THE TRANSFORM OF A DELTA FUNCTION AT THE ORIGIN IS A CONSTANT

Unit impulse



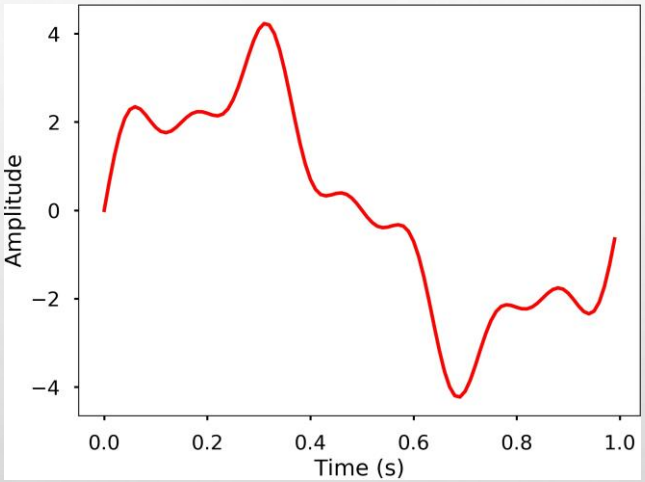
$$F(u) = \frac{1}{N}\sum_{x=0}^{N-1}\delta(x)e^{-i2\pi(ux/N)} = \frac{1}{N}$$

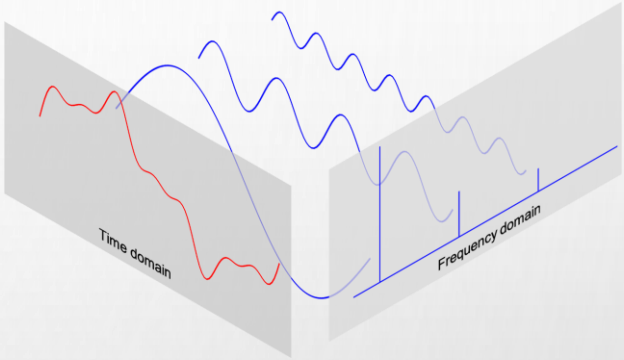
- THE TRANSFORM OF A CONSTANT FUNCTION IS A DC VALUE ONLY.

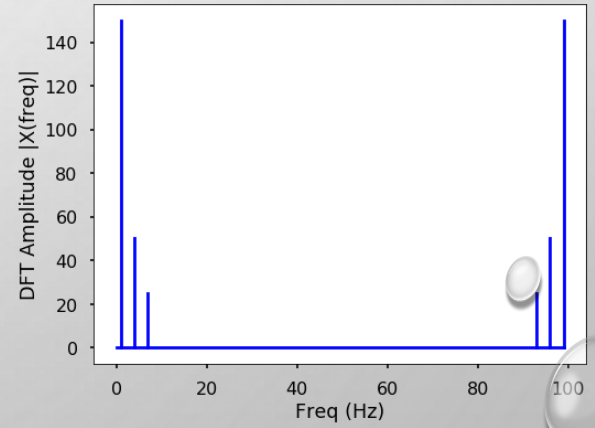


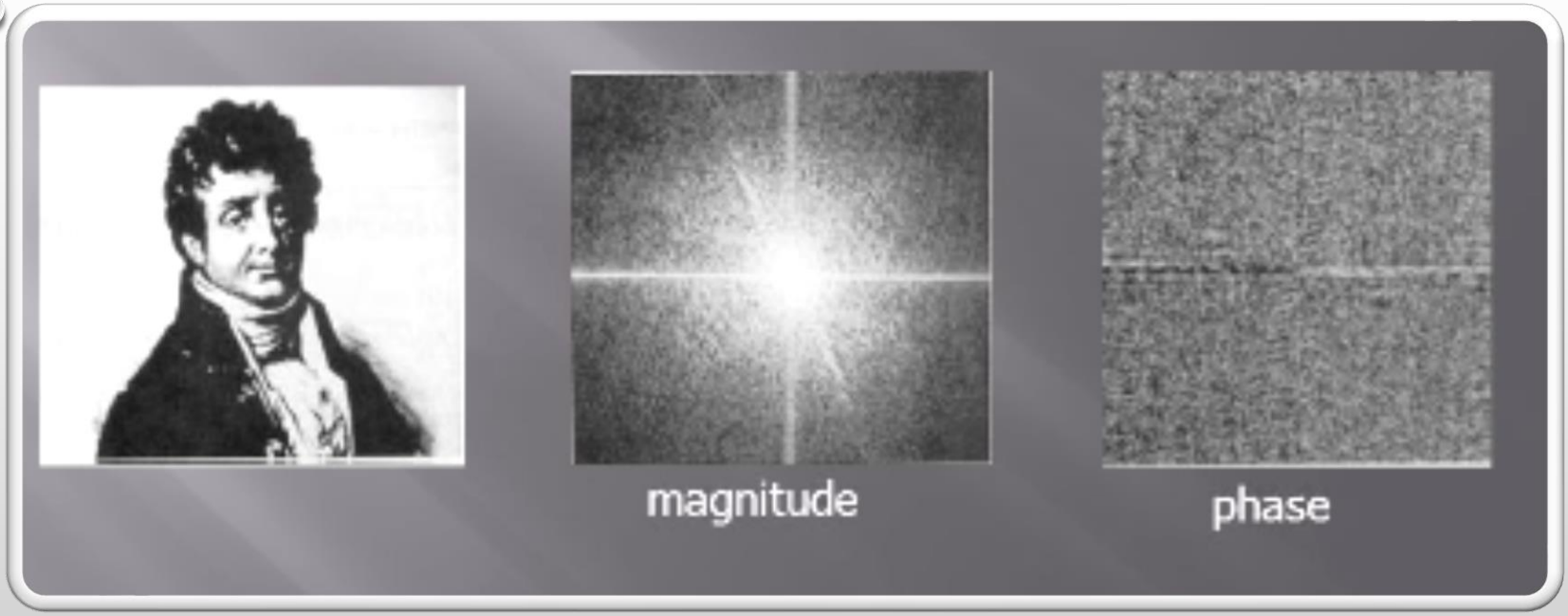
$$f_x = \sum_{u=0}^{N-1}\exp\left[2\pi i\frac{xu}{N}\right]F_u$$

EXAMPLE

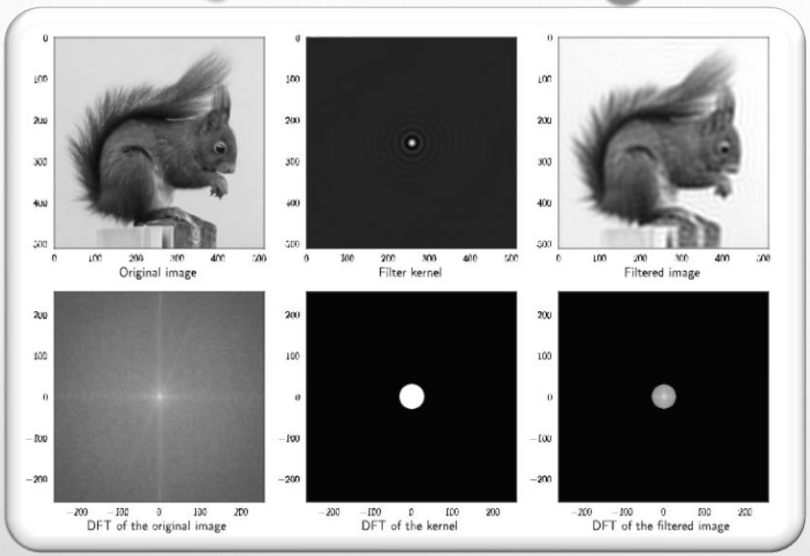

Amplitude
Time (s)
4
2
0
−2
−4
0.0
0.2
0.4
0.6
0.8
1.0


Time domain
Frequency domain


DFT Amplitude |X(freq)|
Freq (Hz)
140
120
100
80
60
40
20
0
0
20
40
60
80
100



2D FFT EXAMPLE



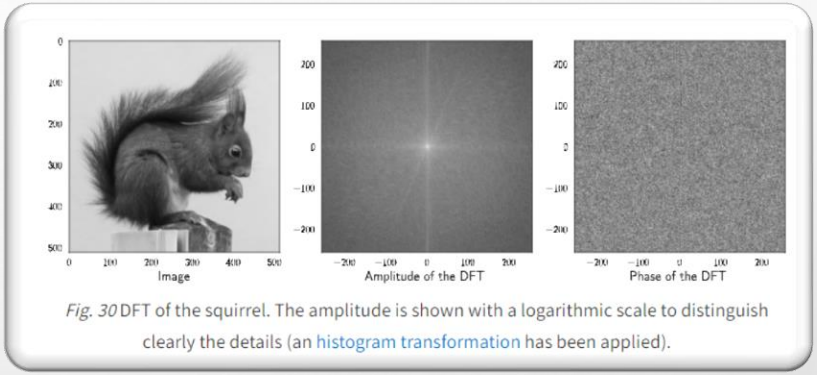


Fig. 30 DFT of the squirrel. The amplitude is shown with a logarithmic scale to distinguish clearly the details (an histogram transformation has been applied).

LOW PASS FILTER EXAMPLE

What is the advantage of carrying filtering in the frequency domain over spatial domain?

The primary reason is that in frequency domain, the process of filtering i.e. convolution simplifies to multiplication. Basically the concept of frequency domain mathematics says that given a function $f(x, y)$ and a kernel $g(x, y)$, the filtered output can be written as $f(x, y) * g(x, y)$ where * represents the convolution operator. Convolution can be explained as placing the kernel over the signal and basically summing up the element-wise products and replacing the anchor value with the sum. On the other hand, in frequency domain, the output is simply given by $F(u, v) \times G(u, v)$ where $F(u, v)$ and $G(u, v)$ are the Fourier transforms of their respective functions. This representation is both computationally and spacewise efficient as both may be represented as sparse matrices as well in most cases.

And hence, we use frequency domain for filtering. Generally though, for separable filters, it makes more sense to just use the separated components one at a time.